POLICY BRIEF

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VIRTUAL REALITY IN EDUCATION

A New “Institution of Connectivity” in our Multi-Order World

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EXECUTIVE SUMMARY

Order in today's world is not solely shaped by a limited number of major powers; rather, it emerges from numerous and diverse actors, encompassing both established and emerging powers, as well as non-state entities.

The contemporary global landscape is at a crossroad, *shifting from established international structures* to a dynamic and complex **multi-order world**.

The existing approach towards global order and governance, rooted in traditional geopolitical norms, falls short in accommodating the emergent dynamics of **shared identities and values** that transcend conventional borders. To surmount this challenge, the brief recommends a decisive course of action: the establishment of "**institutions of connectivity**" to bridge gaps between different orders, notably through the integration of **virtual reality** (VR) **in education**.

This strategic initiative offers a transformative pathway forward. By fostering collaboration and providing immersive learning experiences, it not only addresses the complexities of the multi-order world but also opens doors to a more inclusive, equitable, and interconnected global future.

For *Agenda 2030* to be reached, results and solutions must be delivered quickly and on a large scale. This policy brief asserts the pressing need for immediate action in the face of the insufficiency of current policies to effectively address the nuanced challenges posed by the multi-order system. The urgency lies in the potential repercussions of a delayed response, as the world grapples with composite actors, projected changes, and the imperative for a peaceful transition. The proposal effortlessly integrates with the six signature solutions of the UNDP's new **Strategic Plan 2022–2025** (*Figure 1 and 2*): **the time to adapt and embrace innovative solutions is now**.

Figure 1: UNDP's development offer

Figure 2: UNDP's work summarized in the Strategic Plan by this "3x6x3" framework

EDUCATIONAL CRISIS AND GLOBAL DISRUPTIONS

Education is not merely a means of acquiring knowledge; it is a powerful instrument for fostering understanding, empathy, and collaboration across diverse cultures and nations. A **global education framework** that prioritizes knowledge and skills through equality and organization can address the root causes of **many contemporary challenges** by promoting inclusive and informed perspectives.

Recognized as a **fundamental human right** and a **driver of development**, education has been facing an unprecedented crisis exacerbated by **COVID-19**:

- the closure of schools systems for an average of 141 days globally between February 2020 and February 2022;
- the unequal access to remote learning, and;
- the widening digital divide;

have collectively contributed to an emergency that extends beyond the realms of education, impacting *mental health*, exacerbating *inequalities*, and jeopardizing the *long-term economic stability of nations*.

The causes of such education crisis trace back to the pre-existing learning difficulties that plagued low- and middle-income countries, with **57%** of 10-years-old unable to read and understand age-appropriate texts **before the pandemic**.
The digital divide, compounded by the lack of electricity, devices, and caregiver support, **widened disparities**, particularly affecting vulnerable groups such as children from disadvantaged households, girls, students with disabilities.

The staggering losses extend beyond learning, with a potential lifetime earnings shortfall of US$21 trillion for this generation – equivalent to 17% of today's global GDP.

The urgency of the situation requires immediate and decisive action *not only to recover lost learning*, but to build a more connected, resilient, and inclusive education system. The implications of inaction extend beyond the immediate crisis, posing a threat to the **ambitions of SDGs** and the broader goals of fostering global order, stability, and sustainable development.

DECODING THE CHALLENGES: WHAT IS GOING WRONG?

According to the 2023 Global Education Monitoring (GEM) Report, if nations were moving closer to accomplishing their 2030 national goals:

1. **6 MILLION MORE CHILDREN WOULD BE ENROLLED IN EARLY CHILDHOOD EDUCATION**
2. **THE NUMBER OF CHILDREN, ADOLESCENTS, AND YOUTH ATTENDING SCHOOL WOULD HAVE INCREASED BY 58 MILLION**
3. **THERE WOULD HAVE BEEN AN ADDITIONAL 1.7 MILLION SUITABLY TRAINED PRIMARY SCHOOL TEACHERS**

To accomplish these goals:

- Every year, **1.4 million** kids should sign up for early childhood education.
- Until 2030, a new student must enrol in the system every two seconds.
- Primary completion rates need to **almost triple** annually.

The Report shows how the availability of teaching and learning resources has significantly expanded thanks to **digitalization**. The *National Academic Digital Library* of Ethiopia and the *National Digital Library* of India are two examples. In Bangladesh, more than 600,000 people use the *Teachers Portal*.

Yet, the focus must be on learning outcomes, not on digital inputs.

As a matter of fact, when more than a million computers (and **$200 million**) were given away in Peru without being integrated into the curriculum, **learning did not increase**.

*information and
communications technology

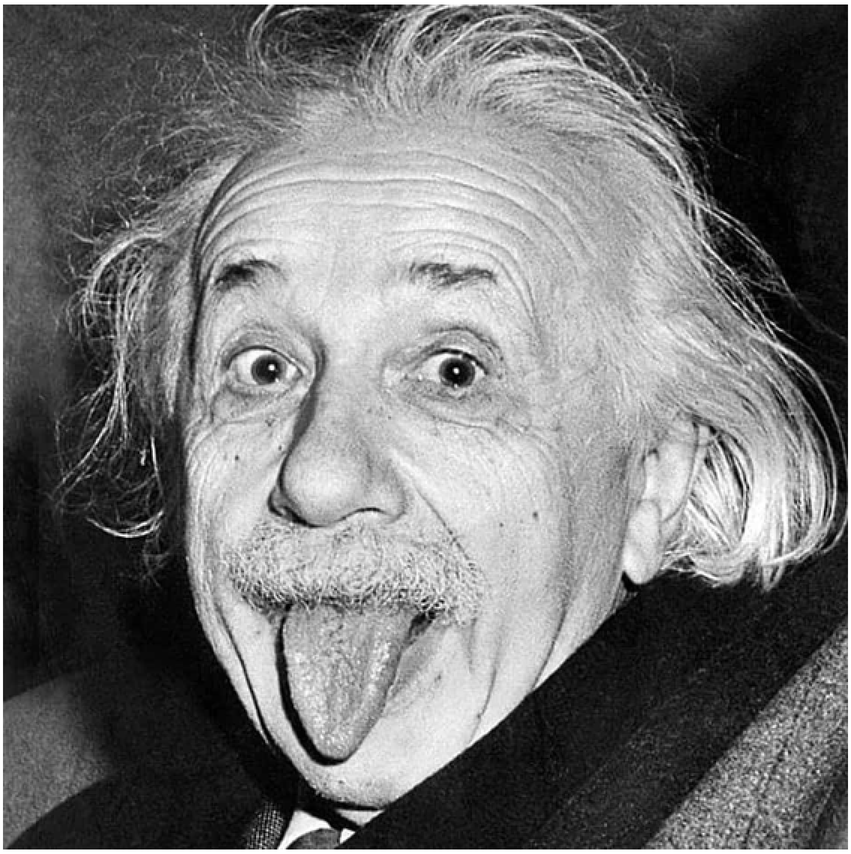
Excessive use of ICT* is in fact negatively correlated with students' achievements, according to large-scale international assessment data, such as the one supplied by the **Programme for International Student Assessment**. Moreover, UNESCO reports showed that just being close to a mobile device can **divert pupils** and **negatively affect their learning**.

Why is technology not helping?

Why are $200 million used without any impact on learning outcomes? Because the investments and efforts need to be Cohesive, Controlled and Concerted. Because mobile devices, computers and remote learning are feasible solutions for unexpected and short-term disruptions, but **not the future of education**. True education should be **immersive**, fostering connection over individualism, while meeting the unique needs of children and students. Moreover, the mentioned digital technologies and investments, unlike VR, l**ack scalability** and fail to promote **interoperability standards** for increased efficiency. While access to meaningful connectivity and **the right to education** are becoming more and more synonymous, access is still **unequal**. The percentage of elementary, lower secondary, and upper secondary schools worldwide with internet access is just 40%, 50%, and 65% respectively. While policies aimed at enhancing learner or school connectivity are implemented in 85% of nations, true inclusivity demands a comprehensive approach. Fortunately, VR learning emerges as a **versatile** solution, capable of operating both with and without the internet, offering pre-loaded or locally stored experiences, and actively addressing obstacles to access. The possible exploration of **solar-powered VR systems** presents a promising avenue for overcoming electricity challenges in isolated regions, ensuring equal access to educational VR experiences. Beyond connectivity, the immersive feature inherent to VR distinguishes it from traditional technologies such as computers and mobiles, potentially reshaping educational outcomes linked to **students' concentration levels on a global scale**.

A CENTURY-LONG DIVIDE

Albert Einstein said, “The only source of knowledge is experience”.



Today, **there is not a single nation on Earth without some sort of educational system**, and for most young people worldwide, the quality or quantity of the education they receive in school will significantly impact their chances of success as adults. The developing world lags developed countries **by around a century** when considering average education years and attainment levels. And unfortunately, the current educational techniques will likely result in this century-long divide persisting into the future. An innovative approach such as immersive VR, emerges as a natural successor in **reshaping education** and bridging the persistent global education gap.

POLICY RECOMMENDATIONS

Immersive classrooms and VR headsets stand out as key tools for incorporating virtual reality into **educational settings**. Immersive classrooms involve projecting images onto interior walls, creating a virtual space within the classroom. The immersive environment encourages **collective learning** and helps introverted or slower-paced students overcome social or attitudinal barriers. On the other hand, VR headsets are compact and require minimal space and equipment. Their portability allows for **easy sanitization and shared use**, enabling multiple VR-based learning programs with **a single investment.**

VR technology is particularly valuable for virtual field trips, offering students a captivating and budget-friendly alternative to traditional travel. It promotes inclusivity by eliminating the need for permission slips, addressing concerns about handicaps, costs, or transportation issues, **ensuring 100% participation and 0% discrimination**. Despite initial costs, the long-term savings on travel, tickets, safety, and time make VR field trips an appealing and financially savvy choice. The safety factor adds to their attractiveness, especially in regions where **security concerns and financial constraints** may impede physical travel. Additionally, the time saved on logistical arrangements and travel can be redirected towards enhancing the overall educational experience.

Beyond classrooms and field trips, the engagement potential of virtual reality extends to exploring diverse environments, **societal challenges**, and **historical sites**. VR provides a dynamic platform for fostering cultural understanding, **combating xenophobia** and stereotypes, and transcending physical boundaries. Virtual field trips, such as **those to the Arctic**, ignite students' interest in **environmental issues**, nurturing awareness and motivating them to care about global challenges. Students can envision themselves as crucial agents capable of driving positive change in the world.

IMPLEMENTATION PLAN

The implementation of a comprehensive staged plan by the UNDP that prioritises **affordability**, **safety**, and **sustainability** can effectively tackle the obstacles hindering the use of VR in education and make it accessible.

00

Pilot Programs and Iterative Refinement

1. Identify a group of **pilot countries** that are representative of different educational, cultural, and economic environments. Consider elements like **infrastructure**, **accessibility**, and **VR integration preparedness**.
2. Provide robust systems for administrators, teachers, and students to provide **feedback** to learn more about the success of VR programmes. To evaluate impact, use **performance indicators**, **interviews**, and **surveys**.

01

Infrastructure Development and Capacity Building

1. Together with the governments and national educational authorities, conduct a comprehensive **needs assessment** to identify the technology capabilities, infrastructure, and unique educational difficulties that the country faces.
2. Create a special **UNDP global fund** named "*UNDP Connects*" to offer assistance to educational institutions and other organisations wishing to purchase VR equipment. This fund can assist in covering the upfront expenses related to buying **software and hardware**.
3. Collaborate with producers of VR hardware to develop educational institutions-only **subsidised programmes** that guarantee access to affordable and reliable VR technologies.
4. Promote the creation of **open-source** virtual reality systems that let organisations share and alter instructional materials. This has the potential to drastically **lower licencing fees** and promote international cooperation among educators.
5. Establish a central repository for educational contents so that institutions may access and exchange them: contents should be free or low-cost.
6. Assure that educational institutions have the resources to set up virtual reality labs, including the technology, software, and **internet access if required**.
7. Cooperate with tech businesses in the private sector to supply VR equipment to schools at a reduced cost or through *sponsorship*. Establish partnerships that cover continuing maintenance, software updates, and support in addition to the original hardware supply.
8. Urge tech companies to **provide grants or enter contests** centred around developing innovative, creative, affordable virtual reality solutions for educational use.
9. Develop and distribute international **ergonomic guidelines** for VR: this will lessen the negative physical effects and health risks linked to extended VR use.

02

Customized Learning Module Development

1. Encourage **global cooperation** between **teachers**, **content producers**, and **VR developers** to build an archive of adaptable VR-based lesson plans and projects.
2. Closely collaborate with national educational authorities to include specially designed VR courses into current curriculum so that they are in line with each country's **educational requirements**.

03

Teacher Support and Professional Development

1. Provide teachers with thorough training programmes that focus on VR integration into the curriculum, efficient use of VR tools, and **best practices** in modifying instruction for immersive environments.
2. Create more **interactive modules** to help teachers use cutting-edge teaching techniques and make sure they can use the technology to accommodate a range of learning styles.
3. Create a **virtual workshop** platform that enables teachers worldwide to collaborate in professional development sessions, *even if they are in different geographical locations*.

04

Student-Centric Integration and Global Connectivity

1. Assist in the progressive implementation of **proficiency-based grouping** in collaboration with educational authorities to provide a seamless shift from conventional grade-based classrooms to **more personalised learning settings**.
2. Use VR-based assessments for **diagnostic testing** to guide **student placement** in level-based classrooms so that instructional strategies can be **dynamically** adjusted.
3. Use VR technology to facilitate **global learning experiences** between students, promoting cross-cultural understanding and global perspectives through **diverse and inclusive initiatives**.

05

Continuous Improvement and Sustainability

1. Establish a thorough evaluation system to gauge the **long-term effects** of tailored, VR-integrated curricula on **global connectivity**, **equality**, and **educational outcomes**.
2. Create **hubs for innovation and research** to investigate new VR applications in education, guaranteeing ongoing enhancement and flexibility to meet *changing learning objectives*.
3. Keep up **capacity-building** initiatives that give educational institutions the tools they need to use VR technology **sustainably** and **creatively**.

FINAL REMARKS

In the comfort of their classrooms, surrounded by familiar faces, students worldwide will now have the opportunity to **exchange experiences** that will profoundly shape their growth, ultimately fostering **unity**. Education serves as a bridge, nurturing mutual understanding and dismantling apprehensions and stereotypes towards the unfamiliar "*others*". A **standardized education**, guided by learning objectives dedicated to **narrowing the disparity between developed and developing nations** and achieving the **2030 Sustainable Development Goals**, empowers students to learn new languages, explore new interests, virtually visit remote locations, and broaden their horizons.

Linguistic barriers will gradually fade, allowing for more resources dedicated to studying regional languages, preserving cultural heritages, and upholding traditions. VR technology creates a platform to share unique experiences with the global community and equips teachers and students with an **extensive repository of knowledge**. The emergence of a **new generation of teachers**, who recognize the value of *human connection*, becomes imperative to ensure the effective use of these new tools.

The UNDP stands uniquely positioned to lead such a transformative initiative, rallying organizations, stakeholders, and governments onto a new trajectory propelled by VR. This journey involves evaluating the potential of VR in education as an "**institution of connectivity**".

Urgent action is imperative to fulfill the commitment made to **millions of children** globally, ensuring that the right to access high-quality education is **not merely a promise**, but a **tangible reality**.

SOURCES RECOMMENDED

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Thank you for your continued support in UNDP efforts to contribute to the SDGs.

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